Prediction of Radar-Derived Quantitative Precipitation Estimation Using Basic Machine Learning Techniques

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Abstract

The advent of artificial intelligence and machine learning within the atmospheric science community has allowed for the continued expansion of boundaries established by the current state of computational power. From emulating climate models with half the normal computing power to detecting features on radar, machine learning has become one of the most essential tools for research in atmospheric science. This study utilizes machine learning to predict precipitation derived from the KLOT NEXRAD radar, which covers the Chicago area. Radar-derived rainfall is known to have issues with accuracy, as radar tends to be biased towards larger drop sizes, making estimations shaky at best for lower-intensity stratiform events with smaller drop sizes. Machine learning algorithms were applied to radar-derived and ground-truth precipitation data, including basic and multiple linear regression in log-log space, as well as a multi-layer perceptron neural network. These algorithms were developed and evaluated using standard statistical performance metrics. Success was achieved when applying the multilayer perceptron on a dataset of ~400 points, with an RMSE of .9303 and correlation (R) of .86. Applying machine learning models to simple tasks such as precipitation estimation allows for the retrieval of patterns or correlations in data that may be difficult to spot with more conventional methods.

Introduction

In recent years, the advent of artificial intelligence (AI) and machine learning (ML) in atmospheric science has presented numerous new research opportunities for the community to explore. In particular, machine learning has become a staple in the realm of prediction through regression techniques, classification, and image recognition, which allows researchers to not only improve the accuracy of various predictions but also automate algorithms to classify specific images based on their visual features. Prominent in atmospheric science is the question of how to effectively estimate precipitation accumulation and rainfall rate from radar in real-time. Radar-derived QPE comes with its own inherent biases, which cause current algorithms to often over- or underestimate precipitation, sometimes by a large amount. Researchers typically use a power law formula known as the Z-R relationship, which directly correlates radar reflectivity (Z) to rainfall rate (R), represented as $Z = aR^b$. The Z-R relationship can assume various forms to address radar's inherent bias toward larger drops, which can lead to precipitation overestimation or underestimation in certain cases. Some commonly used relationships include the National Weather Service's (NWS) deep convection relationship, defined as $Z = 300 * R^{1.4}$, and the standard Marshall-Palmer distribution, defined as $Z = 200 * R^{1.6}$ (Marshall & Palmer et al. 1948). Users can adjust the parameters a and b according to specific needs. The radar's inherent bias toward larger hydrometeors often leads to inaccuracies, causing radar-estimated precipitation to over- or under-represent actual precipitation events. Other inaccuracies arise from beam elevation, anomalous propagation, bright banding, and attenuation, among others (Hunter 1996). To derive the rainfall rate explicitly, the equation is rearranged $R = (Z/A)^{1/b}$. To correct the biases introduced by the Z-R relationship-derived rainfall, the existing network of rain gauges is used to correct the calculated amounts, which are often the data seen on public-facing maps released by the NWS. Since AI/ML has become such a valuable tool for prediction (regression), its integration into radar-derived QPE estimations has been explored extensively. Many different studies have covered the use of regressor-type ML models for the prediction of rainfall from radar (Liu et al. 2024, Huangfu et al. 2024), while other studies have taken a deep learning approach with neural networks (Yo et al. 2021, Li et al. 2023) with good results and significant improvements on the empirical algorithms, increasing accuracy in radar-derived precipitation. The improvement of radar-derived QPE algorithms would enable more accurate measurements, requiring significantly less correction from ground-based rain gauges, currently the primary method of bias correction. The primary focus of this project is to compare radar-derived rainfall rates and reflectivity with instrument-observed ground-truth rainfall to attempt a prediction of rainfall rates from Z using basic machine learning models, where we utilize radar-derived data as predictors and instrument observations as ground truth.

Methodology

A. Radar and the Z-R relationship

In atmospheric science, researchers utilize precipitation radars to detect the location and intensity of precipitation. Each radar is a National Weather Service standard WSR-88D, a 10.5 cm S-band wavelength radar with polarimetric and Doppler capabilities. These capabilities enable the radar to detect storm intensity and motion as well as identify precipitation characteristics, such as scatterer shape, distribution, and uniformity. The WSR-88D radars form the Next Generation Radar (NEXRAD) network, consisting of 160 individual radars strategically located throughout the United States and its outlying territories (NOAA ROC). Given the advanced capabilities of the WSR-88D and its proximity to the areas of interest, it is highly suitable for research purposes. Radar reflectivity (Z) measures the backscattered power from an emitted radar pulse and provides information about storm intensity and size; higher values indicate either larger hydrometeors (such as hail or big drops) or more intense precipitation. Utilized in this study will be the NWS standard Z-R, and a proprietary relationship derived using machine learning techniques.

B. Ground Instruments and the Chicago Micronet

The Chicago micronet is a network of meteorological sensors strategically placed throughout the Chicagoland area. There are eleven different micronet sites to date, each featuring a suite of meteorological instruments, including ambient weather sensors, air quality sensors, lidars, and micro rain radars, which are all managed through an edge computing box known as a WAGGLE node. The micronet sites at the Argonne Testbed for Multiscale Observational Science (ATMOS) and Northeastern Illinois University (NEIU) will be utilized (Fig. 1). On each node, a Vaisala WXT-536, an ambient weather sensor with precipitation measurement capabilities, is installed. The WXT transmits a suite of meteorological data, including barometric pressure, ambient temperature, dew point, relative humidity, wind speed and direction, and accumulated precipitation. The WXT features a heated precipitation sensor that measures the volume of individual raindrops and converts each drop into an electrical signal. Its accuracy is within 5% (excluding wind-induced errors). The WXT will be used as ground truth.

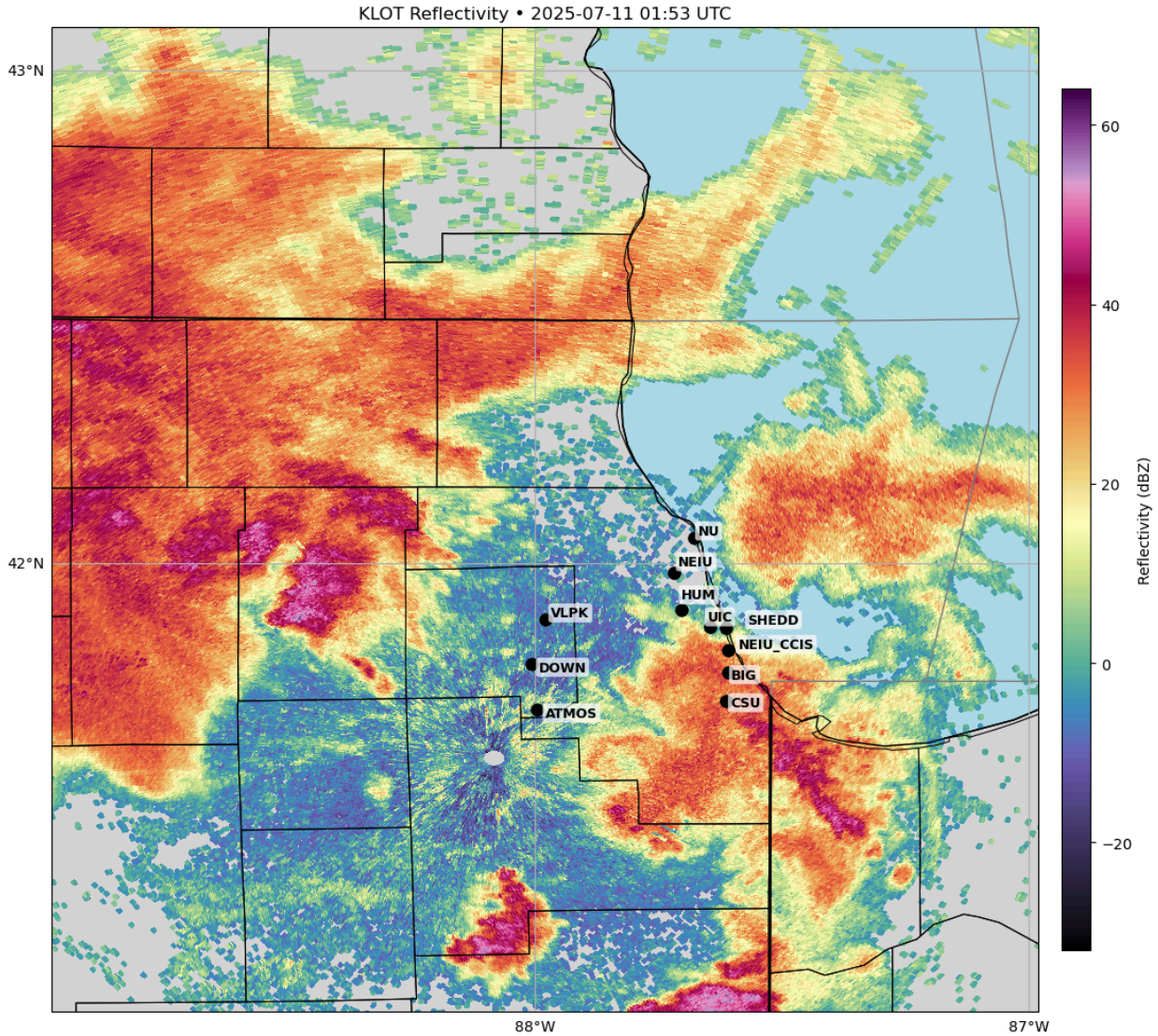

Figure 1: The locations of the current Chicago Micronet sites throughout the Chicagoland area with KLOT radar reflectivity as an overlay.

C. Basic Machine Learning Techniques

AI and machine learning encompass a range of complexities. Ranging from basic models such as ordinary linear regression, more complex ensemble models such as a random forest or gradient-boosted models (find a citation), to multi-million parameter U-Net models and beyond (Ronneberger et al. 2014). When determining the best model type for an application, several key factors must be considered, including dataset size and linearity, data balance, the desired output (regression or classification), and the research question being addressed. In this study, a

regression-type prediction was chosen because we aim to predict a continuous value rather than classify events or images. To derive a Z-R relationship, a linear regression model was selected. Since the variables were non-linear, we chose to use logarithmic space to represent our data, which was achieved by taking the natural log of Z and R. This provides a power law function representing a conversion between Z and R based on the data, effectively establishing a proprietary Z-R relationship. To compare with linear regression, a shallow multi-layer perceptron (MLP) is utilized to predict values of R based on Z and compare them to the instrument values. A GridSearch Cross-Validation was run to determine the optimal hyperparameters for the model (Table 1).

L2 Regularizer	Hidden Layers	Nodes	Activation	Learning Rate
0.01	1	8	RELU	1e-3

Table 1: A table of the best hyperparameters for the multi-layer perceptron model

This neural network was designed to remain relatively straightforward while still handling nonlinear and mildly unbalanced datasets with precision, thereby saving computational power and producing accurate predictions. To evaluate the model's performance, the root mean squared error (RMSE), coefficient of determination (R^2R²), and Pearson correlation coefficient (Rr) will be used.

D. Data cleaning and masking

Given the large number of days used as input for our machine learning models, we prioritized identifying and removing potential outliers and values that could skew the model's output. We downloaded radar files for the target period, extracted reflectivity from the lowest tilt over the chosen sites, and averaged the data over hourly intervals to account for varying scan frequencies across different volume coverage patterns (VCPs). After converting reflectivity to linear Z units, we applied the standard Z-R relationship to estimate hourly rainfall in millimeters per hour. We retrieved WXT data from the GCE cluster at Argonne National Laboratory or queried it using the SAGE data client via Python. From this dataset, we extracted rainfall measurements only when the sensor temperature exceeded 5 degrees Celsius and averaged them over hourly intervals to match the radar data, thereby helping to mitigate sampling issues related to the different temporal resolutions of each dataset. We excluded any hourly periods with rainfall below 0.01 millimeters per hour, as such values typically represented outliers that would

reduce model accuracy. Because the MLP model predicts rainfall directly from reflectivity without relying on a Z-R conversion formula and uses features and targets in different units, we applied a standard scaler to normalize the data before training.

Results

A. Empirical Z-R Relationship

The standard National Weather Service deep convection Z-R relationship is utilized over the timespan of dates to generate an overall conversion from linearized reflectivity to rainfall rate. The empirical Z-R relationship generated by the NWS is meant to account for both stratiform and convective rainfall, but is still known to over- and underestimate quite frequently. The values of R computed from Z were compared to the ground truth WXT R values, netting an RMSE of 1.411 mm/hr, an R^2 of 0.635, and an R of 0.812. The normal Z-R relationship performed reasonably well, given its known biases and the numerous events compiled into the testing data, but suffered from underestimating rain rates during higher-intensity events (Fig. 2).

Figure 2: (a) Linear regression comparing the Z-R predicted rainfall rate and the observed rainfall. (b) Residuals

B. Linear Regression-derived Z-R relationship

The data from NEXRAD and the WXT were cleaned and then inserted into a linear regression model in logarithmic space, which yields a straight-line function in power-law form. The resulting equation would be used as a custom Z-R relationship, explicitly tailored to the provided data. The linear regression (Fig. 3) resulted in a Z-R relationship of $R = 0.03 * Z^{.60}$.

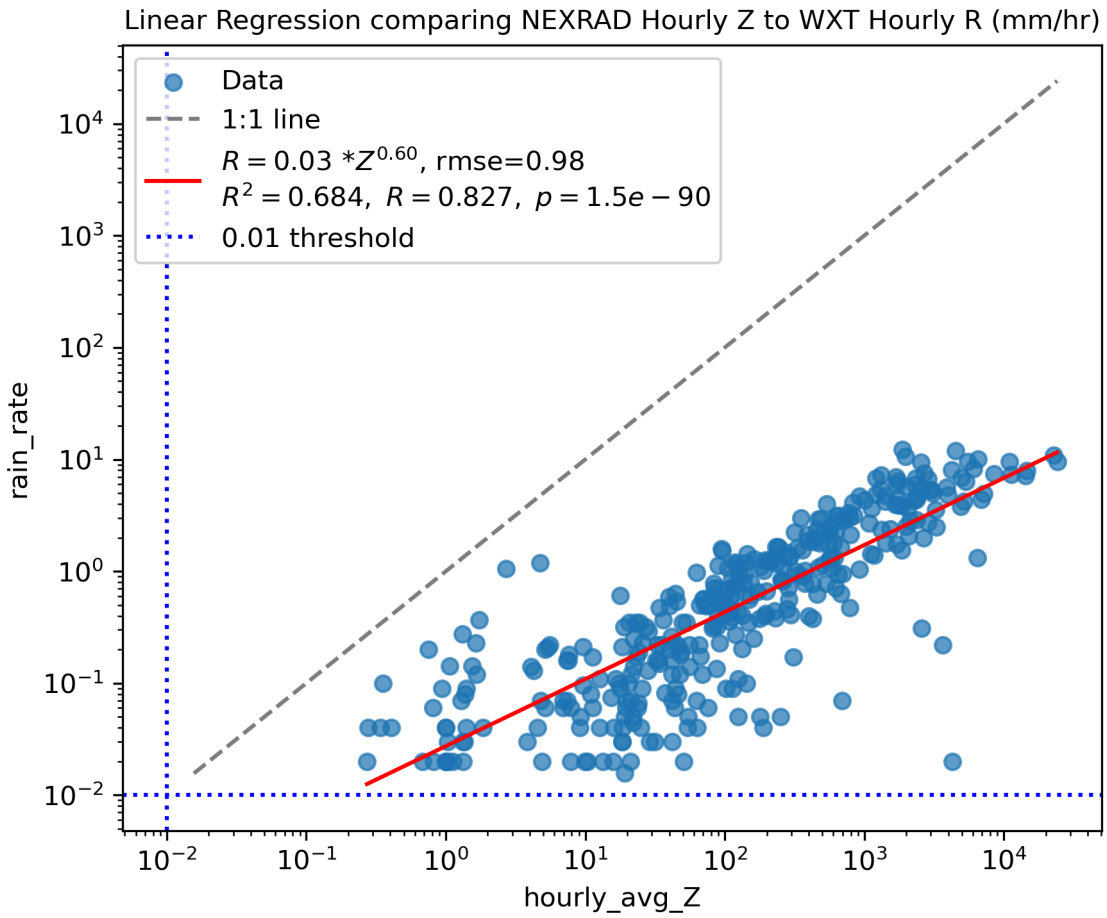

Figure 3: Linear regression derived between NEXRAD hourly Z and WXT hourly R. The resulting equation is the data-tailored Z-R relationship.

Computing R from Z with the new relationship yielded slightly worse performance than the empirical algorithm, with an RMSE of 1.64 mm/hr, a R^2 value of 0.503, and an R of 0.845. The new relationship had significant underestimation issues, particularly for higher-end events (Fig. 4), with residuals peaking around 10 for some of the highest-intensity events.

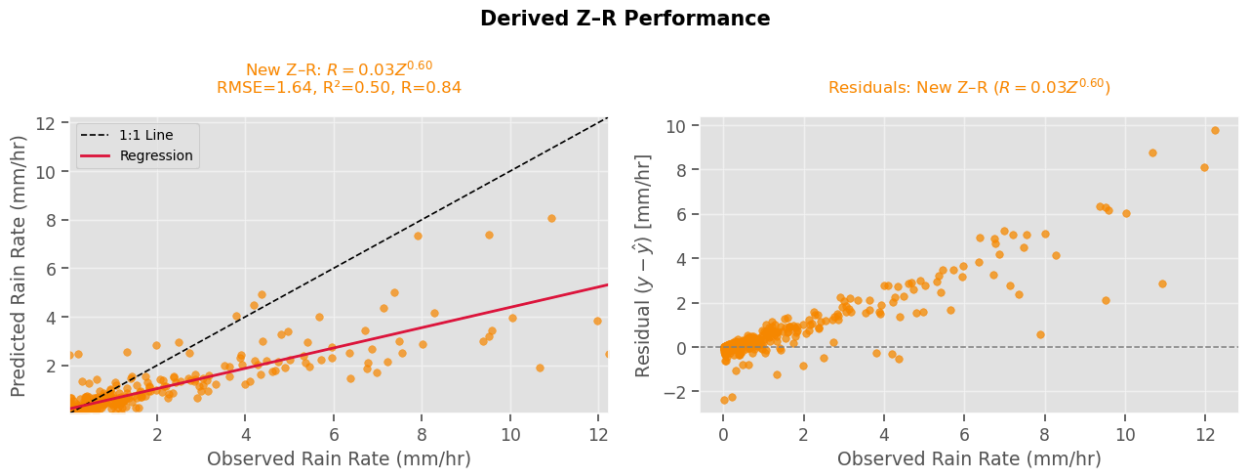

Figure 4: Same as Figure 3, but for the derived Z-R.

C. Multilayer Perceptron Prediction

The same Z and R data were placed into the multilayer perceptron, and the model was run after the data was cleaned and hyperparameter tuning was performed. The training and validation data showcased a model that did not overfit and performed better than both the empirical and derived Z-R relationships, with the MLP model's RMSE being 0.7097 mm/hr less than the empirical and 0.4797 mm/hr less than the derived Z-R, respectively (Fig. 5, Table 2).

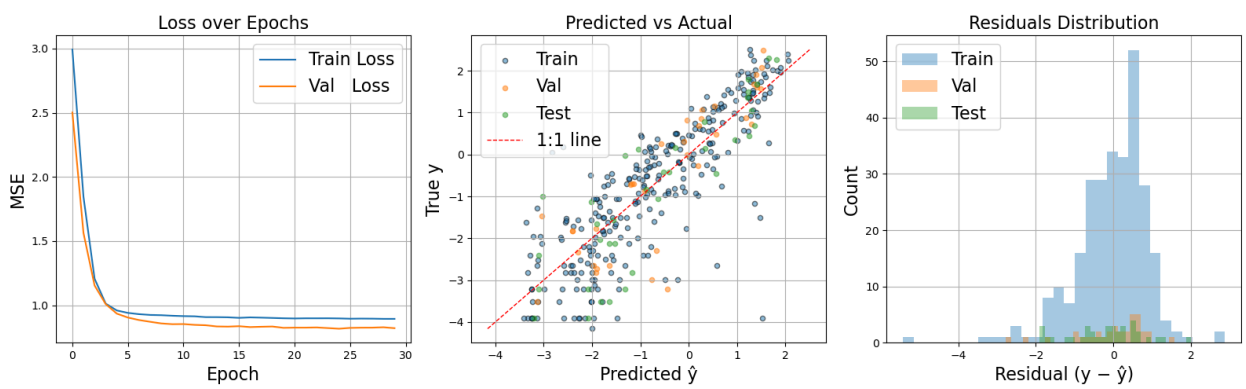

Figure 5: Performance of the multilayer perceptron. (a) The loss curve for training and validation data. (b) The distribution of the points and their predictions. (c) The distribution of residuals for each point in each split of the data.

The model was configured to run for 100 epochs, but early stopping capped the model at 30 epochs as that was a point of diminishing returns. The performance over each split highlights the unbalanced nature of the data. Although the validation data performs better than the training data, the testing data performs slightly worse than both (Table 2), but still markedly better than the linear regression-derived and empirical relationships.

Data Split	MSE	RMSE	R^2	R
Train	0.8644	0.9297	0.7033	0.8386
Validation	0.7912	0.8895	0.7391	0.8597
Test	0.8655	0.9303	0.7430	0.8620

Table 2: Evaluation of model performance for each dataset split.

The five-fold cross-validation confirmed that the model was unable to generalize well, as its performance was slightly biased by the splits. The RMSE had a standard deviation of 0.1391, while R^2 and R had standard deviations of 0.0531 and 0.0312, respectively. This suggests that, despite the dataset's unbalanced nature, the MLP model still outperforms the empirical and derived relationships.

Discussion

A. Linear Regression Performance

Interestingly, the derived Z-R relationship performs less well compared to the empirical relationship, despite being explicitly tailored to the data. This poor performance may stem from the dataset being unbalanced or from the temporal resolution of the averaged data. The unbalanced nature of the data stems from the sporadic occurrence of heavy precipitation events throughout the study period, as well as the short-lived nature of existing heavy precipitation events. This creates a discrepancy between the number of extreme and non-extreme precipitation events, which causes any function that the model derives to be catered towards a dataset consisting mostly of non-extreme events, with extreme events being infrequent outliers. The temporal resolution of the data may also contribute to the model's unfavorable performance. Since we are averaging the data into hourly averages, we are likely not capturing the maximum reflectivity values that occurred in each hour, and we are depriving the model of extra samples of higher-end rainfall, which causes the derived relationship to underperform in accounting for those events.

Attempts were made to address this issue using the Synthetic Minority Oversampling Technique (SMOTE). Due to the limited number of high-intensity samples, duplicate samples were created instead of similar but distinct samples (Fig. 6). To balance the data, data from a larger temporal period with more heavy rain events would be needed.

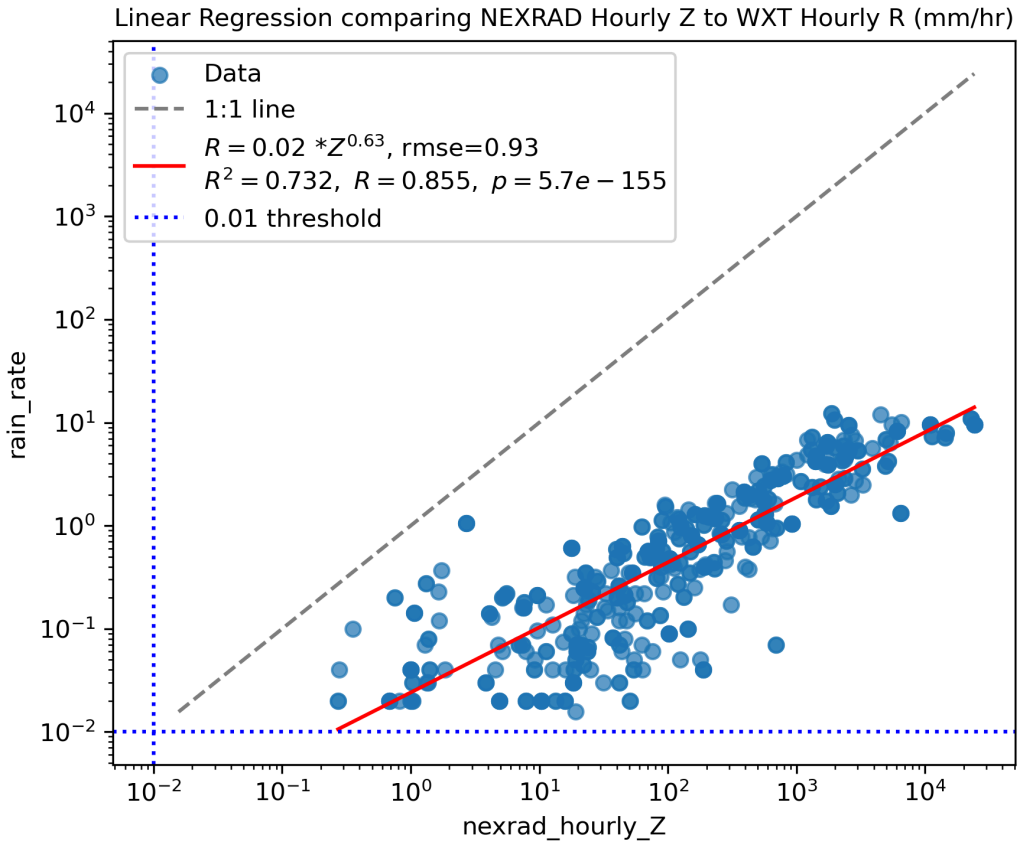

Figure 6. Linear regression-derived Z-R with SMOTE applied. Compared to Fig. 3, the even more significant overlapping (note the darker points) of points and the reduced spread of the data. Instead of adding more natural variability, duplicate points were made.

B. Multilayer Perceptron Performance

The performance of the multilayer perceptron was significantly better than that of the empirical and derived Z-R relationships. The significant performance improvement likely stems from the more advanced capabilities of a neural network compared to those of a linear regression. The MLP model effectively combines multiple linear regressions to capture several relationships, rather than just one, allowing it to minimize RMSE further and deliver roughly 35% and 44% improvements over the empirical and derived algorithms, respectively (Table 2). There is also little benefit to using a linear regression model over an MLP model due to the limited computing resources required to run this model efficiently. Initially, issues with vanishing gradients were present (Fig. 7), and therefore, the decision was made to add L2 regularization to the model to remedy this issue. The complex nature of the data makes it well-suited for an MLP model, and with the approach taken, great results were achieved with minimal complexity and computational draw.

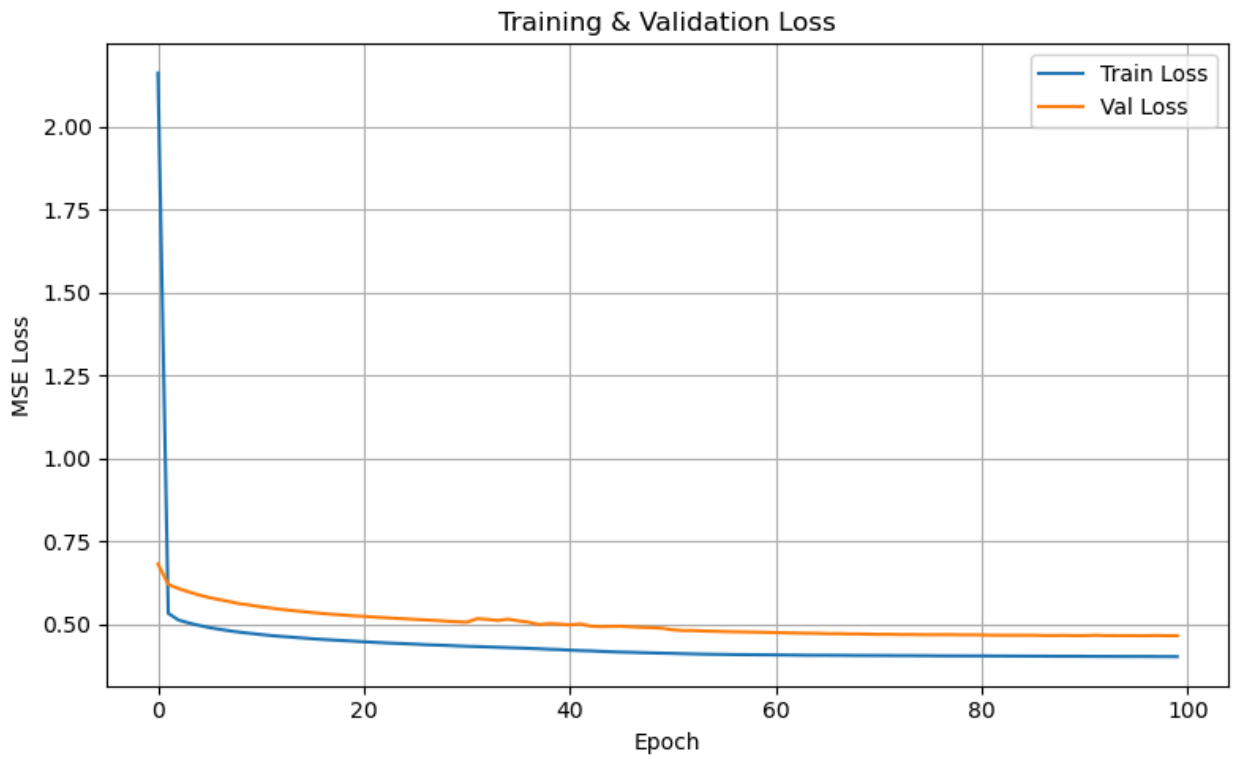

Figure 7: Example of a run where the vanishing gradient issue occurred. The loss curve dropped much more quickly than it should've, which prevents the model's layers from effectively learning. Notice the rapid stagnation of the validation loss as well.

Conclusion

In this study, we employed elementary machine learning methods to enhance the existing algorithms for deriving rainfall rates from radar reflectivity. The machine learning methods used were a basic linear regression and a multilayer perceptron neural network. Data from the WXT-536 sensor at Northeastern Illinois University served as the ground truth, while radar data from the KLOT NEXRAD radar was provided as the input data. It was found that the linear regression performed worse than the empirical Z-R relationship for the data, which is likely due to its simplicity. The MLP model performed significantly better than both the empirical and derived Z-R relationships, with performance improvements exceeding 30%. Despite an unbalanced dataset and suboptimal generalization across data splits, the MLP model still performed the best. Future work will involve implementing this algorithm at various micronet sites in Chicago for instantaneous radar correction at specific points, and possibly developing a more complex model. The ability to instantaneously estimate rainfall rates accurately would serve as a vital tool for forecasting flooding and extreme rainfall in urban areas, such as Chicago, and providing necessary warnings and resources to the areas that are most heavily impacted

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